



ADVENTIST UNIVERSITY OF CENTRAL AFRICA

INTRODUCTION TO BIG DATA

Week1 Python Lab Assignment Report

Name: Mutangana Joseph

Student ID: 29061

Course: Introduction to Big Data

This week's lab introduced me to the foundations of python and showed me how basic programming concept can be used to work with data. I learned how to convert a value stored as text into an integer and how string can be manipulated using method such as **upper()** and slicing.

I also worked with list and dictionaries. With lists, I practiced accessing values by their positions, adding new values, changing existing values, and calculating statistics such as average, highest score, and lowest score. With dictionaries I learned how to organize information using key-value pairs.

For the main part of the lab, I worked with a dataset containing information about 20 students. I represented each student's information using a dictionary and stored the dictionaries inside a list. I extracted student's names, scores, and district into separate lists and use Python to calculate the class average, highest and lowest scores, score range and number of students from particular districts.

One thing that surprised me was how much useful information could be obtained using simple Python operations. For example functions such as **max()**, **min()**, **sum()**, and **len()** made it possible to perform basic analysis without writing complicated code.